Reducing Exposure Before Birth: An Interrupted Time Series Study of Prenatal Exposure to Fetotoxic Medications Under Risk Management Plans in Canada

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Canada's 2009 risk management plan (RMP) framework has not been evaluated for prenatal exposure impact. Conversely, widely used drugs such as nonsteroidal anti-inflammatory drugs (NSAIDs) lack RMPs. We assessed first-trimester exposure to RMP-regulated medications following regulatory interventions and to NSAIDs following safety publications. We conducted interrupted time series analyses using data from the Québec Pregnancy Cohort (1998–2022), including pregnancies among individuals aged 15–45 years with continuous public prescription drug insurance. First-trimester exposure was defined as dispensed prescriptions overlapping or initiated during gestational weeks 0–14. Exposures included RMP-regulated medications (overall, retinoids, and valproic acid) and prescription NSAIDs. RMP dates served as interruption points. Monthly incidence rates were analyzed using segmented Poisson regression to estimate incidence rate ratios (IRRs) with 95% confidence intervals (CIs). Among 559,215 eligible pregnancies, 6,215 and 87,820 involved first-trimester exposure to RMP-regulated medications and NSAIDs, respectively. The Canadian RMP framework was associated with an immediate 45.3% reduction (IRR: 0.547, 95% CI: 0.403–0.743) and a sustained 1.3% monthly decline (IRR: 0.987, 95% CI: 0.981–0.994). Retinoid exposure declined by 12.5% per month post-SMART RMP (IRR: 0.875, 95% CI: 0.772–0.993), with reductions post-iPLEDGE RMP (IRR: 0.949, 95% CI: 0.932–0.966). Valproic acid exposure declined after the strengthened FDA boxed warning (IRR: 0.869, 95% CI: 0.836–0.903) and migraine prophylaxis contraindication (IRR: 0.754, 95% CI: 0.543–1.046). NSAID publications yielded transient effects. RMPs were associated with sustained prenatal exposure reductions, whereas safety publications were associated with transient effects, highlighting the need for reinforced risk-management strategies.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

☑ Medications with teratogenic and fetotoxic potential are used during pregnancy despite known risks. Risk management plans (RMPs) and safety publications have been implemented to reduce prenatal exposure, but evidence on their real-world population-level effectiveness remains limited.

WHAT QUESTION DID THIS STUDY ADDRESS?

☑ This study evaluated whether Canadian RMPs and influential safety publications were associated with changes in first-trimester prenatal exposure to fetotoxic medications over 25 years in Québec.

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

☑ Enforceable RMPs were associated with immediate and sustained reductions in prenatal exposure, whereas safety publications produced only transient effects. Spillover effects from American regulatory programs on Canadian prescribing were also demonstrated.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

☑ These findings underscore that sustained, enforceable risk-minimization strategies may be more effective than safety communications alone, informing the design of future pregnancy drug safety policies globally.

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Medication use during pregnancy is common, with approximately 70% of pregnant individuals requiring at least one prescription during gestation.[1,2] While many therapies are beneficial for maternal health, some medications pose teratogenic risks during the first trimester and fetotoxic risks later in pregnancy.[3,4] Globally, nearly half of all pregnancies are unplanned,[5] thereby increasing the likelihood of inadvertent gestational exposure.[4,6,7] Past drug-related tragedies, most notably thalidomide,[8] prompted regulatory agencies to implement medication risk minimization strategies during pregnancy. In Canada, the formalization of risk management plans (RMPs) by Health Canada in 2009 represented a major advancement in medication safety oversight.[9] RMPs are not always pregnancy related, but measures such as controlled distribution programs, prescriber education, and enhanced safety warnings may nonetheless influence prescribing among women of reproductive age and, consequently, prenatal exposure. Emerging safety signals and scientific publications, whether focused on pregnancy or not, may also modify prescribing behavior during pregnancy.

Despite growing emphasis on structured safety strategies, evidence regarding their real-world impact during pregnancy remains limited. Most prior studies have focused on individual medications or medication use outside pregnancy,[10–12] with limited evaluation of population-level prenatal exposure following regulatory interventions. In addition, assessment of Canadian RMPs can be challenging due to the limited transparency regarding their timing and implementation, in contrast to US Food and Drug Administration's (FDA) Risk Evaluation and Mitigation Strategies (REMS), for which implementation timelines are publicly documented.[13] Therefore, we aimed to evaluate population-level changes in first-trimester prenatal exposure to medications regulated under Canadian RMPs before and after regulatory interventions, and following influential safety publications, using an interrupted time series (ITS) design and 25 years of data from the Québec Pregnancy Cohort (QPC). We also examined exposure to nonsteroidal anti-inflammatory drugs (NSAIDs), which are widely used during pregnancy but not subject to pregnancy-specific RMPs, to assess whether safety publications alone are associated with changes in prenatal exposure.

METHODS

Data sources

The QPC is a population-based cohort with prospective follow-up of pregnancies since 1998 among individuals covered by Québec's public prescription drug insurance plan.[14] The QPC links multiple administrative and hospital databases and population registry data by using Quebec's patient unique health identifiers, including the Régie de l'Assurance Maladie du Québec (RAMQ) for demographic information, medical visits, procedures, diagnostic codes, and outpatient prescription dispensations; the MED-ECHO hospitalization database for admission records, procedures, and gestational age at delivery; and the Institut de la statistique du Québec (ISQ) for vital and birth statistics. The first day of gestation is defined as the first day of the last menstrual period (LMP), as recorded in MED-ECHO or ISQ databases and validated by ultrasounds when available. We followed pregnancies from 1 year before LMP, during pregnancy, and until the end of Quebec's public prescription drug coverage or time of death. The QPC has been previously described and validated for perinatal pharmacoepidemiologic research.[14,15]

Study population

We included pregnancies among individuals aged 15–45 years with continuous public drug insurance coverage from 1 year before LMP and throughout pregnancy. We excluded multiple gestations and isolated chromosomal anomalies due to their associated increases in first-trimester diagnostic testing, and fetal surveillance, which bias exposure ascertainment and denominator comparability (**Figure S2**). Cohort entry was defined as the first day of the LMP, with first-trimester exposure assessed until gestational week 14 for pregnancies occurring up to December 31, 2022.

Risk management plans including medications

We identified medications with documented teratogenic or fetotoxic potential using an internal catalogue developed within our research group and cross-referenced with external teratology sources (**Figure S1**), including the *Williams Obstetrics* (25th edition, Chapter 12, Table 12–1)[16] as well as a teratology-based medication list published by Hashimoto *et al*.[17] This list was further supplemented with confirmed and potential teratogenic medications identified by Sarayani *et al*.[7] We verified each medication in Health Canada's Drug Product Database[18] to determine whether it was subject to a RMP, including Pregnancy Prevention Programs (PPPs) or Controlled Distribution Programs. This process yielded 56 medications regulated under Canadian RMPs, which constituted the primary exposure group (**Table S1**).

Exposures

The primary exposure was first-trimester dispensation of RMP-regulated medications. Retinoids and valproic acid were further examined as distinct subgroups, as they had sufficient exposure frequency to support reliable estimation of level and trend changes in ITS analyses. NSAIDs were also examined separately to assess the impact of influential safety publications in the absence of formal RMPs. In ITS analyses, we defined interruption points using the exact documented implementation dates of risk-minimization strategies and safety signals or publication dates and were assigned to the calendar month in which they occurred, consistent with the monthly time scale of the analysis.

For RMP-regulated medications, we evaluated the introduction of the Canadian RMP framework (2009) and Vanessa's Law (Bill C-17, 2014), which strengthened Health Canada's postmarket oversight, including mandatory reporting of serious adverse drug reactions and by granting the Minister of Health authority to require safety information, postmarket studies, label changes, or product recalls when safety concerns arise.[19]

Retinoids included systemic or topical acitretin, alitretinoin, isotretinoin, and tretinoin. Interventions comprised the revised European PPP (2003) which strengthened and harmonized pregnancy risk-minimization measures for isotretinoin,[20] Health Canada cardiovascular safety communications (2006),[21] the Health Canada Notice of Compliance (NOC) for alitretinoin (Toctino®) (2009) which included an RMP (confirmed by GlaxoSmithKline Canada, personal communication, 2025), the U.S. SMART (2002) and iPLEDGE (2006) programs[22] both designed to prevent fetal exposure to isotretinoin, and iPLEDGE transition to REMS (2010).[23] Since isotretinoin is subject to an active PPP in Canada, one of the few RMPs specifically targeting pregnancy, we examined whether the 2003 European PPP revision was followed by changes in prenatal exposure in Canada.

Valproic acid interventions included an FDA Boxed Warning strengthening (2003),[24] highlighting the teratogenic potential and increased risk of major congenital malformations, and a formal contraindication for migraine prophylaxis during pregnancy (2013).[25]

For NSAIDs, we identified four highly cited safety publications raising clinically meaningful concerns about use during pregnancy. Källén *et al*.[26] reported increased risks of congenital malformations, including cardiac defects and orofacial clefts, associated with first-trimester exposure. Similar associations were reported by Ofori *et al.*[27] based on analyses from our research group. Additional publications highlighted second- and third-trimester fetotoxic risks: Koren *et al.*[28] demonstrated increased risk of ductus arteriosus closure with third-trimester NSAID use, Risser *et al.*[29] summarized risks including ductal closure, prolonged gestation, and bleeding complications, and Nezvalová-Henriksen *et al.*[30] reported associations with adverse perinatal outcomes, including low birthweight, childhood asthma, and maternal bleeding. Although several publications focused on late-pregnancy risks, first-trimester exposure was used as a marker of changes in prescribing following dissemination of safety evidence.

Using outpatient prescription claims, we ascertained first-trimester medication exposure defined as dispensations overlapping or newly initiated during gestational weeks 0–14. Given that medications with pregnancy related RMPs are intended to be avoided entirely during pregnancy, any first-trimester exposure was considered a marker of regulatory effectiveness. Anatomical Therapeutic Chemical (ATC) codes were assigned to each RMP-regulated medication (**Table S1**), and ATC codes beginning with M01A were used to identify NSAIDs. For each calendar month, incidence rates were calculated as the number of exposed pregnancies per 1,000 pregnancy days at risk.

Statistical analysis

We summarized baseline maternal, pregnancy, and healthcare utilization characteristics for pregnancies exposed and unexposed to RMP-regulated medications and NSAIDs (**Table 1**; **Table S2**). Descriptive statistics were reported as counts and percentages for categorical variables and means with standard deviation for continuous variables, with controlled rounding applied as per ISQ privacy requirements.

Preliminary analyses assessed monthly exposure frequency and temporal stability to determine suitability for ITS modeling. Based on these analyses, retinoids and valproic acid, which exhibited stable monthly exposure over time, were examined separately. RMP-regulated retinoids were analyzed as a single class because they are subject to regulatory interventions that may influence prescribing across the retinoid class; grouping also ensured adequate exposure frequencies to avoid sparse-data instability for ITS analyses.

We conducted ITS analyses using segmented regression following the framework described by Bernal *et al.*[31] Monthly incidence of first-trimester prenatal medication exposure was modeled, with changes evaluated at the predefined interruption points. Models included a continuous time variable, defined as the number of calendar months since January 1998. We incorporated interruption points using two parameters: a binary indicator capturing an immediate level change at the time of the interruption and a continuous postinterruption time variable representing changes in trend following the interruption. When multiple interventions applied to a given medication group, separate level and slope parameters were specified for each event. To account for seasonal variation in prescribing, models also incorporated a pair of Fourier terms (sine and cosine with a 12-month period). Generalized linear models with quasi-Poisson distribution and log link were used to estimate baseline trends, immediate level changes, and postintervention slope changes, expressed as incidence rate ratios (IRRs) with 95% confidence intervals (CI). Analyses were conducted using R version 4.5.1 (RStudio, Posit Software).

Ethics statement

Ethical approval for this study was obtained from the Research Ethics Board of the Centre Hospitalier Universitaire Sainte-Justine (reference nos. 1740 and 2976). Data linkage was authorized by the Québec Access to Information Commission (reference no. 1005446-S). All data were anonymized, and no personally identifiable information was accessible. Individual informed consent was not required.

RESULTS

We identified 559,215 pregnancies in the QPC between 1998 and 2022, including 6,215 exposed to RMP-regulated medications during the first trimester. Compared with unexposed pregnancies, those exposed were older (SMD: −0.44, 95% CI: −0.46 to −0.41) and had higher health care utilization (SMD: 0.96, 95% CI: 0.94–0.99). Large imbalances were observed for epilepsy (SMD: 0.47, 95% CI: 0.44 to 0.49), autoimmune disorders (SMD: 0.47, 95% CI: 0.45–0.50), depression or anxiety (SMD: 0.35, 95% CI: 0.32–0.37), and infection (SMD: 0.37, 95% CI: 0.35–0.40). Similar but smaller imbalances were observed among the 87,820 first-trimester NSAID-exposed pregnancies (**Table S2**).

Canadian RMP-regulated medications

As shown in **Figure 1** and **Table S3**, first-trimester exposure to RMP-regulated medications increased by 2.0% per month before the Canadian RMP framework implementation (IRR: 1.020, 95% CI: 1.017–1.024), then decreased immediately (IRR: 0.547, 95% CI: 0.403–0.743) with a sustained decline thereafter (IRR: 0.987, 95% CI: 0.981–0.994). No significant changes followed Vanessa's Law (2014).

Retinoids

Figure 2 and **Table S4** show that first-trimester retinoid exposure increased until the revised European PPP, which attenuated the increasing trend without a significant immediate level change. The subsequent myocardial infarction safety signal was not associated with significant changes in exposure level or trend. Introduction of the alitretinoin NOC was associated with an immediate increase (IRR: 2.159, 95% CI: 1.125–4.145) followed by a reduction in postintervention trend (IRR: 0.951, 95% CI: 0.925–0.977). As shown in **Figure 3** and **Table S5**, implementation of the SMART program (2002) was not associated with an immediate level change but was followed by a reduced postintervention trend (IRR: 0.875, 95% CI: 0.772–0.993). In contrast, iPLEDGE implementation was associated with an increased postintervention trend (IRR: 1.049, 95% CI: 1.010–1.090) followed by a reduction in slope after transition into REMS (IRR: 0.949, 95% CI: 0.932–0.966).

Valproic acid

As shown in **Figure 4** and **Table S6**, valproic acid exposure increased by 14.9% monthly before 2003 (IRR: 1.149, 95% CI: 1.105–1.194). After the strengthened FDA Boxed Warning, postintervention trends declined (IRR: 0.869, 95% CI: 0.836–0.903). The migraine prophylaxis contraindication was associated with a nonsignificant level reduction (IRR: 0.754, 95% CI: 0.543–1.046).

NSAIDs

Figure 5 and **Table S7** show that first-trimester NSAID exposure in Canada was increasing before the 2001 publication by Källén *et al*. reporting an increased risk of congenital malformations associated with early pregnancy exposure (IRR: 1.699, 95% CI: 1.058–2.730). This publication was followed by a significant reduction in postintervention exposure trends (IRR: 0.600, 95% CI: 0.374–0.965). Subsequent publications in 2006 highlighting

Table 1 Baseline maternal and pregnancy characteristics by exposure to identified RMP-regulated medications

Characteristic	Unexposed to RMP medications $N=553,000$ No. (%)	Exposed to RMP medications $N=6,215$ No. (%)	SMD	95% CI
Maternal age (years, mean ± SD)	28.0 (24.0, 33.0)	31.0 (27.0, 36.0)	−0.44	−0.46, −0.41
Gestational duration (weeks, mean ± SD)	37.0 (16.0, 39.0)	37.0 (16.0, 39.0)	−0.05	−0.07, −0.02
Healthcare consultation[a]			0.96	0.94, 0.99
No	300,335 (54.31%)	835 (13.43%)		
Yes	252,665 (45.69%)	5,380 (86.57%)		
General practice visits[a]			0.41	0.39, 0.44
No	282,950 (51.17%)	2,030 (32.64%)		
Yes	270,060 (48.83%)	4,185 (67.36%)		
Emergency visits[a]			0.12	0.10, 0.15
No	543,630 (98.31%)	5,985 (96.31%)		
Yes	9,365 (1.69%)	230 (3.70%)	0.19	0.16, 0.21
Hospitalizations[a]				
No	537,325 (97.18%)	5,790 (93.26%)		
Yes	15,680 (2.84%)	425 (6.84%)		
Emergency hospitalizations[a]			0.22	0.20, 0.25
No	529,140 (95.69%)	5,595 (90.03%)		
Yes	23,860 (4.31%)	620 (9.98%)		
Tobacco use				
No	—[b]	—[b]	0.00	−0.02, 0.03
Yes	—[b]	—[b]		
Alcohol use			0.06	0.04, 0.09
No	552,485 (99.91%)	6,190 (99.60%)		
Yes	520 (0.09%)	25 (0.40%)		
Drug use			0.04	0.01, 0.06
No	552,180 (99.85%)	6,190 (99.60%)		
Yes	815 (0.15%)	20 (0.32%)		
Obesity			0.03	0.01, 0.06
No	551,640 (99.75%)	6,185 (99.52%)		
Yes	1,360 (0.25%)	30 (0.48%)		
Anemia			0.18	0.16, 0.21
No	535,625 (96.85%)	5,775 (92.94%)		
Yes	17,375 (3.14%)	445 (7.16%)		
Asthma			0.24	0.22, 0.27
No	536,520 (97.03%)	5,685 (91.47%)		
Yes	16,480 (2.98%)	530 (8.53%)		
Autoimmune disorder			0.47	0.45, 0.50
No	516,150 (93.34%)	4,790 (77.06%)		
Yes	36,850 (6.66%)	1,425 (22.94%)		
Depression/anxiety			0.35	0.32, 0.37
No	522,515 (94.49%)	5,210 (83.81%)		
Yes	30,485 (5.51%)	1,010 (16.27%)		
Diabetes			0.12	0.09, 0.14
No	548,650 (99.21%)	6,080 (97.83%)		
Yes	4,350 (0.79%)	140 (2.25%)		

(Continued)

Table 1 (Continued)

Characteristic	Unexposed to RMP medications $N = 553{,}000$ No. (%)	Exposed to RMP medications $N = 6{,}215$ No. (%)	SMD	95% CI
Epilepsy			0.47	0.44, 0.49
No	550,025 (99.46%)	5,520 (88.77%)		
Yes	2,975 (0.54%)	695 (11.18%)		
Chronic hypertension			0.20	0.17, 0.22
No	549,460 (99.36%)	6,005 (96.62%)		
Yes	3,540 (0.64%)	215 (3.46%)		
Hyperthyroidism			0.03	0.00, 0.05
No	552,140 (99.84%)	6,195 (99.68%)		
Yes	855 (0.15%)	15 (0.24%)		
Hypothyroidism			0.12	0.10, 0.15
No	549,990 (99.45%)	6,100 (98.15%)		
Yes	3,010 (0.54%)	115 (1.85%)		
HIV			0.24	0.22, 0.27
No	551,630 (99.75%)	5,995 (96.46%)		
Yes	1,370 (0.25%)	220 (3.54%)		
Any infection			0.37	0.35, 0.40
No	449,010 (81.20%)	4,045 (65.07%)		
Yes	103,990 (18.80%)	2,170 (34.93%)		

Controlled rounding can result in total counts that either surpass or fall below the intended sample sizes for each group.Abbreviations: CI, confidence interval; SD, standard deviation; SMD, standardized mean difference. [a]Healthcare utilization during the first trimester of gestation, defined as the number of general practitioner and specialist visits, emergency department visits, and hospitalizations, extracted from the Régie de l'assurance maladie du Québec (RAMQ) administrative claim databases. [b]The specific value cannot be disclosed due to granularity.

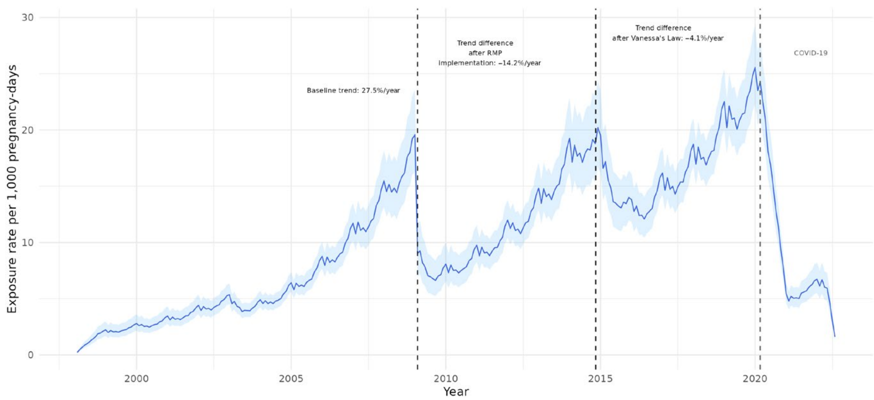

Figure 1 Interrupted time series of prenatal exposure to risk-management plan (RMP)-regulated medications in Canada, 1998–2022 monthly incidence rates of prenatal exposure to medications covered by Canadian Risk Management Plans from 1998 to 2022. The solid line represents model-fitted exposure rates, and the shaded region indicates 95% confidence intervals. For interpretability, monthly slope estimates were transformed into yearly percentage changes by multiplying coefficients by 12 and exponentiating the results. Vertical dashed lines mark the timing of major Canadian interventions, including the introduction of the national RMP policy in 2009 and the implementation of Vanessa's Law in 2014. The gray vertical line marks the onset of the COVID-19 pandemic. The *x*-axis displays calendar years, and the *y*-axis indicates exposure rates per 1,000 pregnancy days.

fetotoxic risks associated with late-pregnancy NSAID use, including premature closure of the ductus arteriosus, were associated with a significant immediate decrease in exposure (IRR: 0.765, 95% CI: 0.614–0.951). The 2009 publication by Risser *et al.* was not associated with a significant immediate change (IRR: 1.216, 95% CI: 0.958–1.545). In contrast, the 2013 study by Nezvalová-Henriksen *et al.* was associated with a significant immediate decrease in exposure (IRR: 0.794, 95% CI: 0.663–0.952), with no meaningful change in the subsequent trend.

DISCUSSION

In this population-based study of Québec pregnancies, we found that first-trimester exposure to medications with teratogenic or fetotoxic potential changed following regulatory actions and

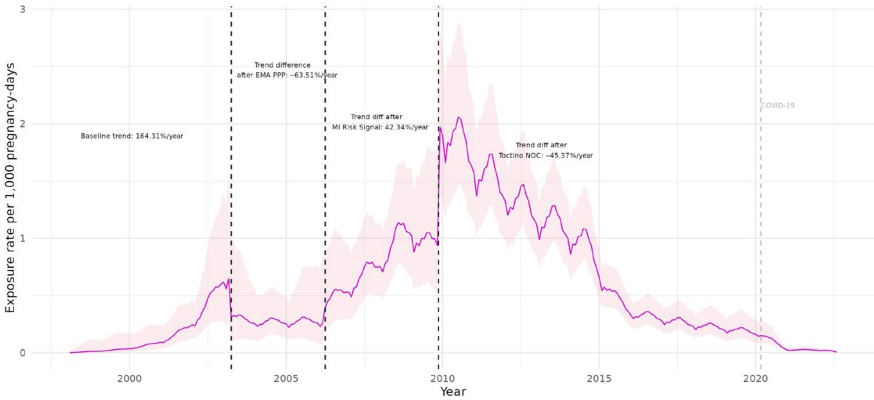

Figure 2 Interrupted time series of prenatal retinoid exposure following European and Canadian risk-management interventions, 1998–2022 monthly incidence rates of prenatal retinoid exposure from 1998 to 2022. The solid line represents model-fitted exposure rates, and the shaded region indicates 95% confidence intervals. The plot includes vertical dashed lines indicating the European Pregnancy Prevention Program (2003), the myocardial infarction risk communication (2006), and the Canadian Notice of Compliance for alitretinoin (2009). Model-based predictions are shown with a magenta line, and uncertainty is represented by a light pink ribbon. The gray vertical line marks the onset of the COVID-19 pandemic. The *x*-axis displays calendar years, and the *y*-axis indicates exposure rates per 1,000 pregnancy days.

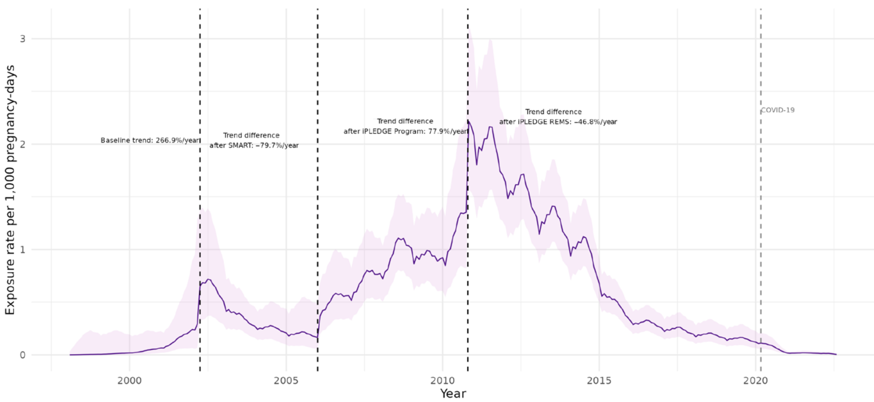

Figure 3 Interrupted time series of prenatal retinoid exposure following US risk-management interventions, 1998–2022 monthly incidence rates of prenatal retinoid exposure from 1998 to 2022 in relation to major US risk-management interventions. The solid line represents model-fitted exposure rates, and the shaded region indicates 95% confidence intervals. Vertical dashed lines denote the introduction of the SMART program in 2002, the iPLEDGE program in 2006, and the iPLEDGE REMS conversion in 2010. The gray vertical line marks the onset of the COVID-19 pandemic. The *x*-axis displays calendar years, and the *y*-axis indicates exposure rates per 1,000 pregnancy days.

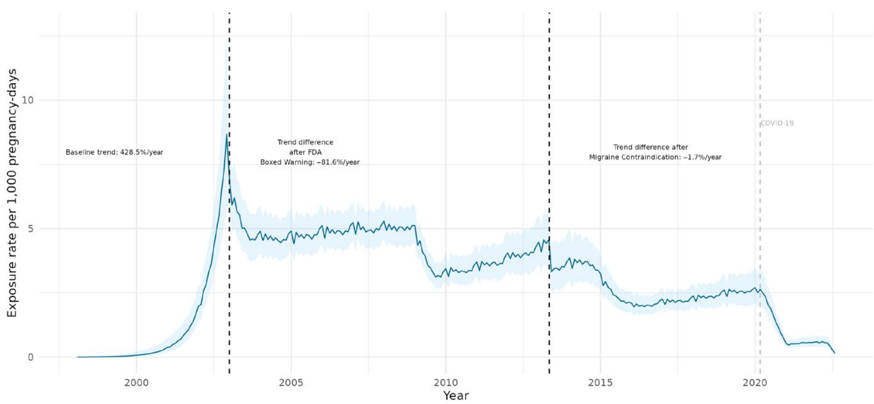

Figure 4 Interrupted time series of prenatal valproic acid exposure following US risk-management interventions, 1998–2022 monthly incidence rates of prenatal valproic acid exposure from 1998 to 2022. The solid line represents model-fitted exposure rates, and the shaded region indicates 95% confidence intervals. Vertical dashed lines identify two major US interventions: the strengthened Food Drug Administration boxed warning in 2003 and the migraine-specific pregnancy contraindication in 2013. The gray vertical line marks the onset of the COVID-19 pandemic. The *x*-axis displays calendar years, and the *y*-axis indicates exposure rates per 1,000 pregnancy days.

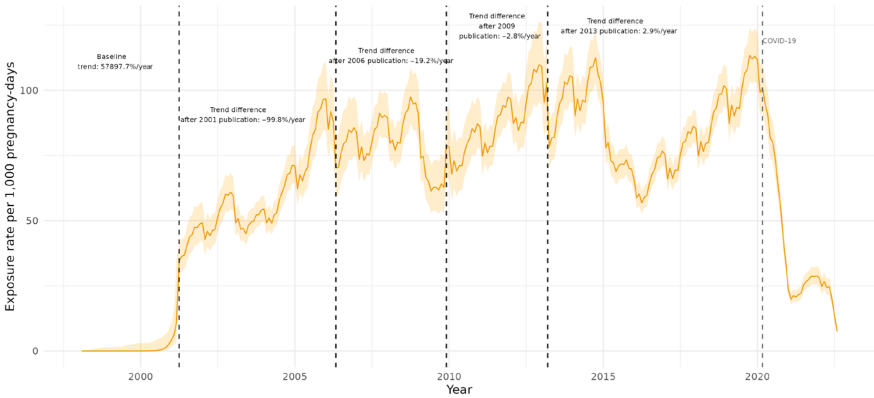

Figure 5 Interrupted time series of prenatal nonsteroidal anti-inflammatory drugs (NSAID) exposure following influential scientific publications, 1998–2022 Monthly fitted incidence rates of prenatal NSAID exposure between 1998 and 2022. Vertical dashed lines correspond to the timing of four highly cited publications addressing NSAID safety during pregnancy (2001, 2006, 2009, and 2013). The solid line represents model-fitted exposure rates, and the shaded region indicates 95% confidence intervals. The gray vertical line marks the onset of the COVID-19 pandemic. The *x*-axis displays calendar years, and the *y*-axis indicates exposure rates per 1,000 pregnancy days.

influential scientific publications. Examining medications with contrasting clinical roles during pregnancy, from contraindicated therapies such as retinoids to clinically necessary treatments such as valproic acid, provided insight into how both intervention design and therapeutic context influence the effectiveness and durability of safety interventions. The introduction of the Canadian RMP framework was associated with a pronounced and durable decline in exposure with an immediate reduction of 45.3% followed by continued decreases.

Among retinoids, implementation of the SMART program was followed by a 12.5% monthly decline, with further reductions after reinforcement of iPLEDGE under the REMS framework (5.1% monthly decline). Our study design does not permit definitive attribution of these trends to direct US spillover. However, the temporal alignment between US regulatory programs and changes in Canadian prescribing patterns may indicate cross-border influences, such as information sharing among clinicians, pharmaceutical industry implementation of risk-minimization measures across North American markets, or parallel evolution of clinical awareness in response to shared safety evidence. In contrast, prior studies reported no reduction with SMART and only partial reductions after iPLEDGE, with exposed pregnancies also persisting post-iPLEDGE REMS.[32–34] Differences in healthcare systems and access to pregnancy planning programs between the United States and Quebec may partly explain these discrepancies.[35,36] Prescribing patterns may also contribute, as dermatologists are the main prescribers of isotretinoin in the United States,[37] whereas general practitioners also prescribe in Canada[38] and may be more reticent to prescribe to women of reproductive age after safety signals. Consistent with this, early retinoid pregnancy-prevention efforts were associated with modest or inconsistent changes in exposure: the 2003 European PPP revision was followed by a limited reduction (8.1% monthly decline), whereas initial iPLEDGE implementation coincided with a short-term increase (4.9% monthly increase) before subsequent REMS reinforcement was associated with sustained declines.

For valproic acid, the strengthened FDA Boxed Warning was associated with a 13.1% monthly reduction and the migraine prophylaxis contraindication with a 24.6% immediate decrease. Prior studies similarly reported declines in valproic acid use among individuals of childbearing potential following strengthened risk-minimization measures.[39–41] Although exposure decreased for both retinoids and valproic acid after regulatory interventions, reductions were larger and more consistent for retinoids.

In contrast, interventions lacking enforceable prescribing restrictions were associated with only transient or incomplete reductions despite short-term statistical significance. Vanessa's Law, aimed at strengthening postmarket drug safety oversight in Canada, was not associated with additional or sustained reductions in first-trimester exposure. A comparable pattern was observed for NSAIDs. Safety publications were followed by an immediate exposure decrease of 23.5% (Koren *et al*., 2006) and short-lived trend shifts, including a 40.0% reduction in monthly trend after Källén *et al*. (2001); however, these effects were not sustained. This aligns with evidence that even stringent pregnancy-prevention strategies do not fully eliminate fetal exposure.[33,34] Despite long-standing pregnancy-prevention efforts, prenatal retinoid exposure was not eliminated, and prior evaluations found no significant reduction in isotretinoin fetal exposures with iPLEDGE before its transition to REMS compared with SMART.[33] Persistent isotretinoin-exposed pregnancies have been reported in Québec, underscoring the difficulty of fully preventing exposure despite established safeguards.[42] Retinoids are contraindicated in pregnancy and used primarily for nonlife-threatening dermatologic conditions,[43,44] continued prenatal exposure remains clinically concerning.

The transient effects observed for NSAID publications should be interpreted considering important differences in use patterns and availability.[45] NSAIDs are widely available over-the-counter, whereas the studied RMP-regulated medications require prescriptions. Our analysis captured only prescription NSAIDs, underestimating total exposure.[45] NSAIDs are commonly used without specialist oversight, which may reduce the impact of safety publications on prescribing, whereas RMP-regulated medications are recognized as higher-risk, which may make prescribers more responsive to formal programs. These differences suggest safety strategies may need tailoring to medication characteristics. One possible explanation for these differences is that pregnancy-prevention programs differ in enforceability and integration into clinical care. Programs such as SMART and iPLEDGE require mandatory prescriber registration, patient consent, and repeated verification steps that may influence prescribing behavior.[22,23] Programs relying primarily on recommendations or educational measures, without systematic checkpoints or broad prescriber engagement, may have more limited impact, particularly where prescriptions are frequently issued by nonspecialists.

Taken together, these findings are consistent with prior evidence that reinforced regulatory interventions that directly influence prescribing or dispensing were associated with greater reductions in prenatal exposure compared with those relying primarily on safety communications or postmarket oversight.[46–48] Finally, strategies implemented outside Canada and safety signals not specific to pregnancy were nonetheless followed by decreases in first-trimester exposure, which may reflect multiple mechanisms including cross-border information sharing through clinician networks, shared literature, and international regulatory harmonization. However, durability varied over time, with exposure re-increasing a few years postintervention, indicating that continued reinforcement may be required.

Strengths and limitations

To our knowledge this is the first population-based study to evaluate Canadian and international pregnancy-related safety interventions across multiple medication classes in Québec. Use of the QPC, with prospectively collected population-level data, validated gestational dating, and complete outpatient prescription capture,[15] enabled precise assessment of first-trimester exposure over time. The availability of 25 years of follow-up permitted evaluation of both immediate and long-term intervention effects. The ITS design strengthened causal inference by distinguishing intervention-associated changes from underlying temporal trends. The pregnancy-specific cohort structure, anchored on the validated first day of the last menstrual period,[49] ensured accurate alignment of interventions with gestational timing and minimized exposure misclassification.[50]

This study has limitations. First, our findings should be interpreted considering potential temporal population changes and co-interventions. Disease prevalence, treatment practices, and pregnancy planning practices may have evolved over 25 years, influencing prescribing independent of regulatory interventions. Additionally, concurrent influences may have contributed to observed trends, such as evolving clinical awareness through professional networks, and gradual diffusion of safety evidence. While our ITS design accounts for underlying secular trends, this design cannot fully separate regulatory intervention effects from these concurrent changes.[50] Second, the interventions examined represent strategies with varying enforceability. Formal regulatory frameworks (Canada's RMP framework and Vanessa's Law) established requirements for manufacturers to implement risk-minimization measures. Structured pregnancy prevention programs (SMART and iPLEDGE) imposed requirements including prescriber registration, patient consent, and verification steps. In contrast, safety publications relied on evidence dissemination without systematic enforcement mechanisms. These design differences likely contributed to observed variation in effectiveness and durability, consistent with prior evidence that regulatory interventions with enforceable requirements are more effective than those relying primarily on safety communications.[46–48] Third, administrative data capture dispensed prescriptions rather than intake. However, validation studies in Québec pregnant populations show high concordance between dispensing records and self-reported use, with positive and negative predictive values exceeding 85–95% for most medications.[15] Misclassification of medication exposure is therefore unlikely to affect observed population-level trends. Fourth, ITS analyses require prespecified interruption dates and cannot capture gradual or retrospectively recognized changes in clinical practice. Nevertheless, ITS remains among the strongest quasi-experimental designs for policy evaluation.[51] Our analyses modeled interventions as discrete interruption points. However, the actual impact of some interventions, particularly safety publications, may reflect gradual awareness of evidence. This may explain transient effects, as clinician awareness may have increased gradually then waned without reinforcement. Similarly, pregnancy prevention program implementation may have varied across clinical settings. The ITS design captures average population-level effects but cannot fully characterize RMP implementation heterogeneity or pace of changes in practices. Fifth, our analyses describe population-level trends and cannot identify which prescribers, patients, or health system factors contributed to observed changes, limiting inference about underlying mechanisms. Sixth, pregnant RAMQ beneficiaries tend to have lower socioeconomic status but are comparable to privately insured individuals in comorbidity, medication use, and health care utilization,[52] supporting generalizability beyond the publicly insured population and to other Canadian provinces. Finally, our definition of first-trimester exposure included prescriptions dispensed before pregnancy overlapping with weeks 0–14, and prescriptions newly initiated during this period. The implications differ by RMP type. For medications with pregnancy prevention programs requiring contraception (e.g., isotretinoin), both scenarios reflect potential RMP implementation challenges, including contraceptive failure, non-compliance, or unplanned pregnancies (for overlapping exposures) and prescribing before pregnancy recognition (for newly initiated exposures). For other RMP medications without pregnancy-specific programs, overlapping exposures may partially reflect unavoidable exposures in individuals on chronic therapy.

CONCLUSION

First-trimester exposure to medications with documented fetal risks changed following regulatory interventions and the dissemination of scientific evidence, with effects varying in magnitude and durability across medication classes. Structured, enforceable risk-minimization measures, such as the Canadian RMP framework, were associated with the most sustained reductions in exposure. However, for several interventions, initial reductions were not sustained and exposure resurged over time. These findings may inform the design of pregnancy safety policies by underscoring that sustained reinforcement, rather than one-time regulatory actions, may be important for achieving durable reductions.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

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CONFLICT OF INTEREST

The authors declared no competing interests for this work.

AUTHOR CONTRIBUTIONS

N.S.K., O.S., L.F.L., V.T., and A.B. wrote the manuscript; N.S.K. and A.B. designed the research; N.S.K. performed the research; N.S.K., O.S., L.F.L., V.T., and A.B. analyzed the data.

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